INDEPENDENT NUMERICAL STUDY · REVIEWED 25 OCTOBER 2026

> **Qualified reproduction — final review complete.** The damping trends are supported; universal convergence, uniformly accurate shocks, and PM superiority are not established. Original science and failures preserved. Final verification · Acceptance and provenance record

32	3,269	32 / 32
paired phase realizations	trajectories · 14 retained stops	implementation tests passed

Independent reproduction of Perona–Malik relaxation filtering for Burgers turbulence

Abstract

Maulik and San (2016) proposed a Perona–Malik (PM) relaxation filter to suppress numerical oscillations and grid-cutoff energy accumulation while preserving Burgers-turbulence spectra. We independently implemented the recoverable equations and completed a paired, 32-realization study across seven resolutions, three conductivity kernels, and a selective-filter comparator. Of 3,269 trajectories, 3,255 completed; 14 under-resolved unfiltered trajectories stopped at a stability guard after the common spectral observation time. Every declared ensemble has all 32 spectra at that time. Increasing PM controls consistently reduced cutoff energy at the tested discretizations. Coarse sine-wave oscillations also decreased, but smoothing introduced shock broadening and smooth-region error. An investigator-defined scaling gate on wavenumbers 20–128 failed even for the reference solution; this does not falsify the paper's broader qualitative scaling claim. Representative timestep refinements leave some production results unqualified, and no comparator pair satisfied the prescribed damping match. The evidence therefore supports specific damping trends, not exact author-curve reproduction, uniform shock accuracy, universal parameter optimality, or PM superiority.

1. Research question and scope

The original question was whether gradient-dependent diffusion, applied once after each physical timestep, could act both as a shock-capturing filter and as a subgrid-scale (SGS) closure. Maulik and San's §4 and Figs. 2–10 report reduced ringing, reduced high-wavenumber accumulation, and improved preservation of approximately k^{-2} scaling relative to selective filtering; §5 proposes wider applicability [1].

We ask which of these observations survive an independent implementation with paired initial conditions, explicit numerical checks, and frozen diagnostics. All numerical results and linked result figures below are **our independent calculations, not author raw data**. Missing author seeds, timestep, and implementation conventions preclude exact replication. The original pilot remains separate from the new production ensemble. The governing commitments are the pilot protocol, stage-two protocol, and additional-time protocol.

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2. Source-based methods

2.1 Evolution and filtering

Following [1, §2, Eqs. (1)–(9)], we solve periodic viscous Burgers flow on unique nodes $x_j = 2\pi j/N$, with $\nu = 5\times10^{-4}$:

$$u_t = -\frac{1}{2}\partial_x(u^2) + \nu\partial_{xx}u, \qquad F(u) = -\frac{1}{2}D_1(u^2) + \nu D_2 u.$$

The discrete conservative nonlinearity is not interchangeable with $-uD_1u$. With $h = 2\pi/N$ and indices interpreted periodically, the sixth-order compact operators satisfy

$$\begin{aligned}\tfrac{1}{3}f'_{j-1} + f'_j + \tfrac{1}{3}f'_{j+1} &= \tfrac{14}{9}\frac{f_{j+1}-f_{j-1}}{2h} + \tfrac{1}{9}\frac{f_{j+2}-f_{j-2}}{4h},\\ \tfrac{2}{11}f''_{j-1} + f''_j + \tfrac{2}{11}f''_{j+1} &= \tfrac{12}{11}\frac{f_{j+1}-2f_j+f_{j-1}}{h^2} + \tfrac{3}{11}\frac{f_{j+2}-2f_j+f_{j-2}}{4h^2}.\end{aligned}$$

These are [1, Eqs. (5)–(6)], inverted through their rational Fourier symbols rather than the paper's indicated tridiagonal algorithm. This is **FFT inversion of compact finite differences, not spectral differentiation**. SSP-RK3 advances F according to Eq. (9). No dealiasing, clipping, upwinding, or extra baseline stabilization was introduced.

For the completed RK field v, one simultaneous conservative PM update implements [1, §3, Eq. (12)]:

$$u_i^{n+1} = v_i + \frac{\kappa\Delta t}{2h^2}\left[(c_i + c_{i+1})(v_{i+1}-v_i) - (c_{i-1}+c_i)(v_i - v_{i-1})\right].$$

With $q_i = (D_1 v_i/K)^2$, the three conductivities are

$$c_1 = (1+q_i)^{-1}, \qquad c_2 = (1+q_i/3)^{-2}, \qquad c_3 = e^{-q_i/2},$$

as in Eqs. (13)–(15). Conductivity uses a freshly computed compact gradient of v, a declared interpretation of the unspecified gradient timing/stencil. The combined post-step method has first-order splitting effects; RK3 alone does not imply third-order filtered trajectories.

The comparator applies Eqs. (16)–(17) once per completed step, with symmetric coefficients $(d_0, d_1, d_2, d_3, d_4) = (35/128, -7/32, 7/64, -1/32, 1/256)$. Its Fourier multiplier is $1 - \sigma \sin^8(\theta/2)$. Therefore, increasing σ increases attenuation, contrary to one source sentence. Fixed σ applied more frequently changes physical-time damping: timestep halving is **algorithm-frequency sensitivity**, not convergence of a fixed physical closure.

2.2 Initialization and source ambiguities

The sine experiment starts from $u(x, 0) = \sin x$ and is evaluated at $t = 2.5$ [1, §4, Fig. 2]. Turbulence uses [1, Eqs. (18)–(23)]

$$E_0(k) = \frac{2}{3\sqrt{\pi}\ k_0^5} k^4 e^{-(k/k_0)^2}, \quad k_0 = 10, \qquad \widehat{u}_k(0) = \sqrt{2E_0(k)}\, e^{i\phi_k}.$$

Positive-mode phases are uniform; negative coefficients are conjugates. We use $\widehat{u} = \mathrm{FFT}(u)/N$, zero initial DC/Nyquist coefficients, and no realization-dependent rescaling. Plotted positive modes retain the undoubled two-sided density $E(k) = |\widehat{u}_k|^2/2$; total energy includes both signs.

Equations take precedence over contradictory prose: $\kappa = \Delta\tau/\Delta t$; the phase multiplies the coefficient, not its modulus in Eq. (21); and Eq. (18) peaks at $\sqrt{2}\ k_0$, not k_0. The source's approximate sine shock-formation time is not imposed as a verification condition: viscous flow has no true discontinuity, and inviscid breaking would occur at $t = 1$. Fig. 2 does not identify its kernel, so all three were tested. Sensitivity snapshots at $t = 0.05$ are an explicit assumption. Source interpretations and recovered legends are documented in methods, figure findings, and higher-resolution findings.

2.3 Paired design and diagnostics

Exactly 32 PCG64 seeds, **160504593–160504624**, generate saved positive-mode phase tables shared across grids and methods. Spectra are computed per realization before averaging. Grid studies use $N = 512, 1024, 2048, 4096, 8192, 16384, 32768$ through $t = 0.1$; coarse sweeps stop at 0.05.

Experiment	Controls at N512, N1024, N2048
PM1/2 K sweeps, source Figs. 4/6	K = 50, 100, 125, 150, 200; κ = 0.0015
PM3 K sweep, Fig. 8	K = 150, 300, 500, **750**, 850 at N512; **700 replaces 750** at N1024/2048; κ = 0.0015
All κ sweeps, Figs. 5/7/9	κ = 0.0005, 0.001, 0.0015, 0.002, 0.0025; K = 125
Selective, Fig. 10	σ = 1/3, 1/2, 2/3
Sine, Fig. 2	K = 200, κ = 0.0015; all kernels and unfiltered controls

The cutoff band is $[\lfloor 0.7k_c \rfloor, \lfloor 0.95k_c \rfloor]$, $k_c = N/2$. We retain positive excess and deficit separately, logarithmic spectral RMS error over $20 \le k \le \lfloor 0.95k_c \rfloor$, low-mode energy error, and sampled/projected field errors. Spectral molecular dissipation $\nu \sum_k k^2 |\widehat{u}_k|^2$, compact viscous loss, finite-cadence energy loss, and post-filter removal are distinct diagnostics; the author's dissipation estimator is unknown.

The investigator-defined useful-SGS gate requires at least 50% cutoff-excess reduction, 20% log-error reduction, and eight PM/DNS bin ratios within [0.8, 1.2] on **20–128**. The reference must first have slope in [−2.2, −1.8] and compensated-bin coefficient of variation (CV) ≤0.2. The slope fits log arithmetic-bin energy against log geometric-bin center in eight logarithmically spaced bins. These endpoints and tolerances are **not supplied by the paper**.

Uncertainty uses 2,000 paired realization-ID bootstrap samples, not Fourier-bin resampling. Intervals describe phase sampling only, **not numerical-error bounds**. Nested first-8/16/32 comparisons test stability, not independent replication. A full-ensemble mean requires all 32 members; otherwise histories contain gaps rather than survivor averages.

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3. Execution and numerical qualification

3.1 Completed coverage and resource accounting

The validated audit counts **3,269 terminal trajectories: 3,255 complete and 14 guard stops**, comprising 3,452 immutable segments and 4,343,614 accepted steps.

Family	Complete	Guard-stopped
DNS timestep/cadence checks	3	0
Seven turbulent grid ensembles	210	14
PM sweeps	2,688	0
Selective sweeps	288	0
Representative turbulence refinements/frequency check	43	0
Sine references	3	0
Coarse sine production and halving	16	0
Separately authorized additional time levels	4	0

All **100 ensemble groups, or 3,200 members**, contain valid $t = 0.05$ fields. The 14 stops are exclusively N512 unfiltered under-resolved DNS (UDNS), at $t = 0.0563$–0.0963; only 18/32 reach 0.1. Their terminal fields are finite. The next attempted actual-velocity stability rate exceeds 0.8; late energy growth is separately observed and is not repaired. Neither a guard stop alone nor that discrete growth falsifies the paper.

The ledger audit confirms **3,834.562955 s (63.91 min)** charged simulation/output wall time against a 5,400-s cap, including the full 240-s interrupted reservation. There are 205 closed reservations: 204 normal chunks and that interruption. The added diagnostics cost 10.305021 s. The final resource audit records 338,112,532 result-file bytes (322.45 MiB), below 1 GiB. Maximum manifest-recorded simulation RSS is 68,943,872 bytes (65.75 MiB), below 2 GiB; this is a saved measurement, not an independently monitored peak. Analysis/testing time is outside the simulation charge. The infrastructure amendment changed checkpoint scheduling, not the science.

3.2 Verification and reference convergence

The independent evidence review reran the existing suite: **32 tests passed**, including dense cyclic-operator checks, smooth-mode order, Cole–Hopf verification, filter transfer, and checkpoint lineage (test log). No tests or simulations were rerun for this write-up. Exhaustive saved-field checks reproduce every spectrum exactly, with maximum Parseval residual 8.88×10^{-16}; regenerated initial coefficients agree within 1.18×10^{-16}. These checks do not replay every evolution update.

At N32768, representative unfiltered timestep levels $d, d/2, d/4$, $d = 4.5015253 \times 10^{-6}$, give successive spectral log-RMS differences 1.303×10^{-6} and 1.616×10^{-7}. Time/cadence gates pass. N16384→32768 ensemble changes include energy 5.14×10^{-8} relatively, molecular-peak height 0.0812%, and spectral log-RMS 0.001049. Nevertheless, four individual peak-time comparisons miss the 1% gate by one 0.0005 output interval: seeds 160504599, 160504609, 160504613, and 160504615. Ensemble agreement does not erase those misses (qualification; independent checks).

3.3 Production timestep limitations

Production coarse turbulence uses $\Delta t = 10^{-4}$, subject to the protocol's compact-symbol bound; coarse sine uses 5×10^{-4}. Actual-velocity guards and $\kappa \Delta t / h^2 \leq 0.5$ are enforced without adaptive rescue. Temporal gates require energy and peak height/time changes below 0.5% and spectral log-RMS below log(1.02), excluding reference energies below 10^{-12} of their maximum; sine adds a DNS-normalized field-change gate of 0.5%. Representative turbulence comparisons pass 9/14 base-to-half and 13/14 final-halving gates. The remaining PM3 miss is retained exactly: **0.0198099548 > log(1.02) = 0.0198026273**.

Four subsequently authorized refinements are **separate diagnostics, not replacement production ensemble members** (analysis).

N2048 diagnostic	Latest spectral log-RMS	Latest temporal gate	Production→finest log-RMS
Turbulence PM3, K125, κ0.0025; added dt=0.0000125	0.008758	pass	0.065738, fail
Sine PM1; added dt=0.000125	0.038941	fail	0.122311, fail
Sine PM2; added dt=0.000125	0.040466	fail	0.126802, fail
Sine PM3; added dt=0.000125	0.041353	fail	0.129300, fail

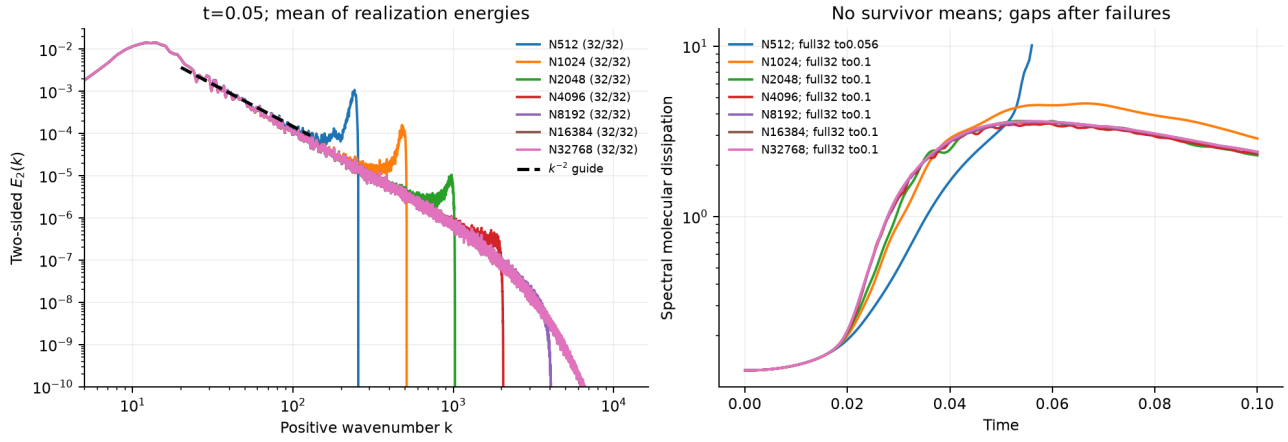
For turbulence PM3, the production-to-finest cutoff-energy shift is 14.79% of the same-seed production κ0.002→0.0025 contrast. Large damping persists, but the lower control was not equivalently refined, and this does not certify all contrasts. All three refined sine cases still fail spectral and molecular-peak gates despite small field changes. Sine reference space/time checks and coarse UDNS halving pass. Fixed-σ=0.5 selective halving instead changes the N2048 spectrum by log-RMS **1.201987**, highlighting the missing author timestep's importance (all refinements).

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4. Results

4.1 Turbulence: damping with spectral costs

Fig. 3 · Matched32-member grid study



Independent paired grid spectra and molecular dissipation histories

Figure A. Independent counterpart of source Fig. 3. Left: mean realization spectra at t=0.05, with all 32 members at every grid. Right: molecular dissipation histories; N512 is not averaged over survivors after losses of completeness. The dashed k^{-2} line is a guide, not a passed scaling test.

The DNS ensemble molecular-dissipation peak is **3.6210 at t=0.056**, with phase-bootstrap intervals [3.4594, 3.7784] for height and [0.0535, 0.0585] for time. This supports a peak near 0.05, not exact timing agreement with [1, §4, Fig. 3].

On the frozen 20–128 interval, the fitted slope is **−2.32972890**, interval **[−2.35456, −2.30311]**, and compensated CV is 0.17550. The CV passes; the slope fails. First-8/16/32 slopes are −2.34872/−2.32896/−2.32973, while the 16→32 full-band log-RMS difference remains 0.22475 (turbulence summary; interval figure). Thus no useful-SGS conjunction passes. This measures neither inertial-range length nor every possible scaling interval; it does not establish absence of an inertial range or disprove the paper's qualitative claim.

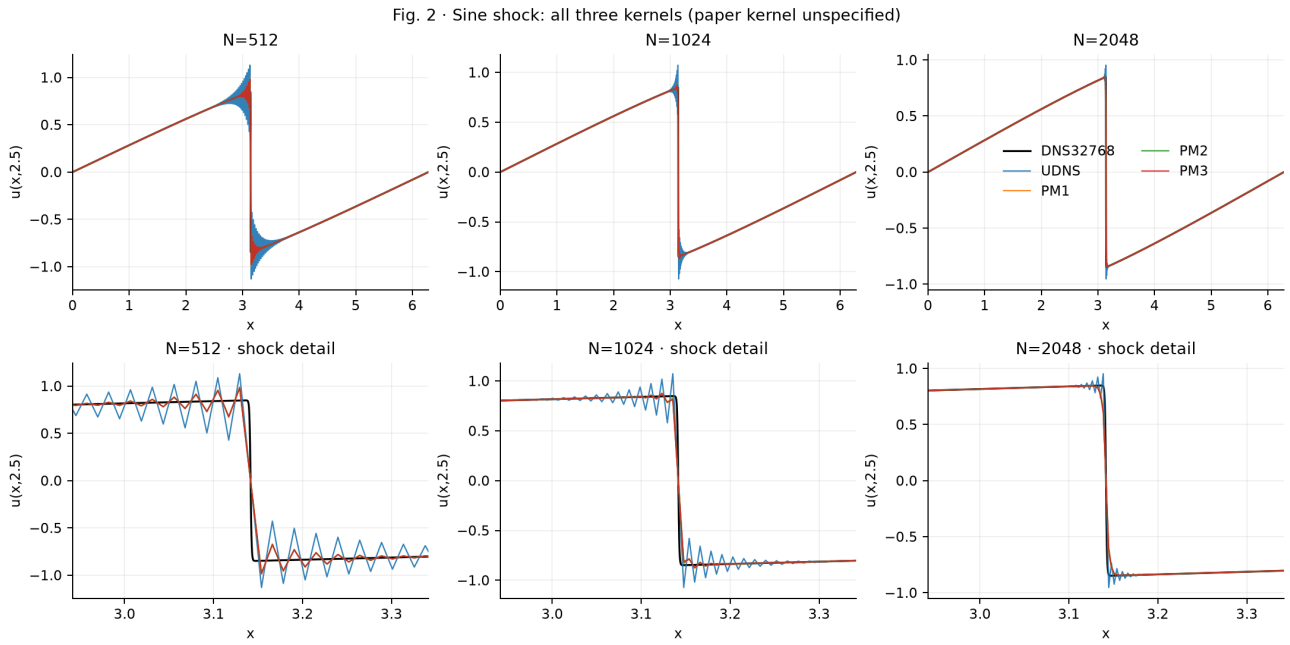
At the tested discretizations, **all 72 adjacent-control contrasts reduce cutoff energy**, with pointwise, unadjusted 95% paired phase-bootstrap intervals excluding zero. These are not simultaneous 95% coverage of all contrasts or uncertainty intervals for a selected optimum.

N	UDNS log error of mean spectrum	Default PM1 excess reduction, mean-spectrum metric	PM1 log-error reduction, mean-spectrum metric	PM1 reduction in mean realization-level log error [95% CI]
512	1.3680	62.25%	29.48%	24.16% [21.93, 26.72]
1024	1.1054	73.19%	42.77%	30.24% [27.89, 32.59]
2048	0.7624	80.57%	61.80%	28.57% [23.82, 32.76]

Default PM1 means K125, κ0.0015. A nonlinear metric of the ensemble-mean spectrum differs from the mean of per-realization metrics; the last two columns must not be interchanged. Evidence: production metrics, summary, and independently checked N512, N1024, N2048 groups.

At κ0.0025, PM1 mean-spectrum log-error reductions reach 45.43%, 63.67%, and 71.26%; this is not a global optimum. Damping also increases mean low-mode absolute-energy error at N1024 by 0.001999 [0.001409, 0.002597] and N2048 by 0.003938 [0.003700, 0.004186] for default PM1. Selective σ0.5 eliminates cutoff excess but produces deficits 0.001444/0.0006485/0.0002574, nearly the reference cutoff energy. **Zero** declared PM/selective pairs match remaining cutoff-band energy within 10%. Generic superiority is therefore inconclusive, not established by unmatched error comparisons (tradeoff figure).

4.2 Sine flow: oscillation reduction is not uniform accuracy



Independent sine profiles and shock-region details at production timesteps

Figure B. Independent counterpart of source Fig. 2 at t=2.5: UDNS, all three PM kernels, and N32768 DNS. These are unchanged production-timestep profiles, not added-refinement results. Lower panels expose ringing and broadening concealed by whole-domain views.

N	UDNS sampled-reference RMS	PM1 / PM2 / PM3 sampled-reference RMS	UDNS width	PM1 / PM2 / PM3 widths
512	0.058414	0.019454 / 0.018936 / 0.018622	ambiguous	0.015946 / 0.016041 / 0.016100
1024	0.022674	0.003582 / 0.003527 / 0.003562	ambiguous	0.009442 / 0.009599 / 0.009708
2048	0.006251	0.006439 / 0.007290 / 0.007853	0.004136	0.006098 / 0.007027 / 0.007602

Production values; these deterministic field metrics are not ensemble averages. DNS width is 0.002327. Independent evidence: sine audit.

Widths use fixed DNS plateau values at $\pi \pm 0.2$, unique descending 10–90 crossings, and linear interpolation; multiple crossings are explicitly ambiguous. They are sampled-profile diagnostics, not independently resolved continuum thicknesses: N512 PM widths span only about 1.3 cells. All location proxies equal π, testing only this grid-aligned shock.

At N512, PM lowers the outside-shock second-difference oscillation proxy by 95.31–95.48%, but widths remain about 6.9 times DNS. At N1024, the proxy decreases by about 99.89% while outside-shock field error increases from 0.0001820 to approximately 0.0002173. At N2048, PM introduces smooth-region error near 0.0002171 against UDNS approximately 4.1×10^{-11}.

At the added finest timestep, PM sampled errors are 0.00618781/0.00703019/0.00759216 versus UDNS 0.00625073. **PM1's small ordering changes with timestep**; its projected-reference error remains worse (0.008348 versus 0.003325). PM2/3 sampled errors remain worse at **all three tested timestep levels**. Finest PM widths, 0.006016/0.006705/0.007338, remain broader than UDNS and DNS; even PM1 spans only about two cells. Uniformly sharp, accurate shock capture is not established.

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5. Figure-by-figure outcome and interpretation

Each link below opens our numerical counterpart, not a digitized author curve.

Source figure [1]	Independent outcome
1: kernels	Specified conductivity limits and gradient dependence supported.
2: sine	Coarse ringing reduced; uniform accuracy/better width unsupported.
3: grid study	Reference spectral agreement and near-0.05 peak supported; fixed slope gate fails; four individual peak-time misses.
4: PM1 K	Increasing-control damping supported; full SGS gate unmet.
5: PM1 κ	Damping supported; global optimum unestablished.
6: PM2 K	Damping supported; full SGS gate unmet.
7: PM2 κ	Damping supported with numerical/low-mode qualifications.
8: PM3 K	Damping supported with grid-specific 750/700 control retained.
9: PM3 κ	Damping supported; original temporal miss not erased by added-level pass.
10: selective	Cutoff damping supported; deficit and unmatched controls prevent superiority inference.

The reproduced contribution is a **bounded qualitative damping trend**, accompanied by measurable spectral and shock-profile costs. Failure of a stronger investigator-defined gate does not negate that contribution. Conversely, successful smoothing cannot establish useful SGS behavior, continuum accuracy, or optimal

parameters. Three-dimensional generalization and dynamic parameter selection in [1, §5] were not attempted. Implementation timings are not author timings; computing compact gradients also prevents equating a three-point filter stencil with negligible total cost.

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6. Review status, limitations, and reproducibility

Both independent reviews conclude **ready_for_qualified_report**, with no substantive unresolved numerical/source P1 blocker: methods review and evidence review. The parent accepted both reviews as supporting qualified reporting, not full paper confirmation. The sine comparison helper's omission of its field criterion from the composite flag remains a future validation-code fix. It has no effect on present outcomes: the auditor independently checked every actual field gate, all passed, and PM spectral/peak failures remain. No protected code or evidence was changed.

The evidence review independently verified all 7,780 inventoried scientific artifact hashes and 24 code/report hashes. That inventory is a historical snapshot, not universal hash coverage of later files. HANDOFF hash drift was confirmed as parent-only orchestration updates; the older exception does not authenticate those later bytes. A separate current-hash authorization is parent-owned and planned after the final handoff update, not claimed complete here.

Reproduction should begin with the saved evidence, not overwrite it. From the workspace root, inspect the following map:

Purpose	Artifact
Primary equations and captions	paper text
Frozen plan and exact expanded controls	stage protocol, plan.json
Environment and dependencies	ENVIRONMENT.md, requirements lock
Phase tables, raw immutable segments	phases/, runs/
Configuration/code lineage and charges	identity-v2.json, ledger/
Aggregates, errors, uncertainty	analysis/, report-data.json
Separate refinements and inventory	extra-time-levels/, final inventory
Executable workflow and audit	stage README, command record, independent audit scripts/results

The existing `.venv` supplies the recorded environment. For separately authorized reruns, use a fresh/copied workspace, numerical threads set to one, and the stage README's sequential initialization, bounded execution, validation, aggregation, and plotting commands. Outputs deliberately refuse overwrite; recovery commands are

historical, not instructions to rerun against this completed archive. No Git repository/ref exists. No new simulation, installation, threshold adjustment, or source-system change was performed during authoring.

Exact author matching remains limited by missing seeds, timestep, Fourier/gradient conventions, and dissipation definition. Representative convergence cannot certify every production control; bootstrap intervals cannot absorb discretization uncertainty; and unmatched selective controls cannot settle relative closure quality. These are limits on interpretation, not reasons to discard the supported damping observations.

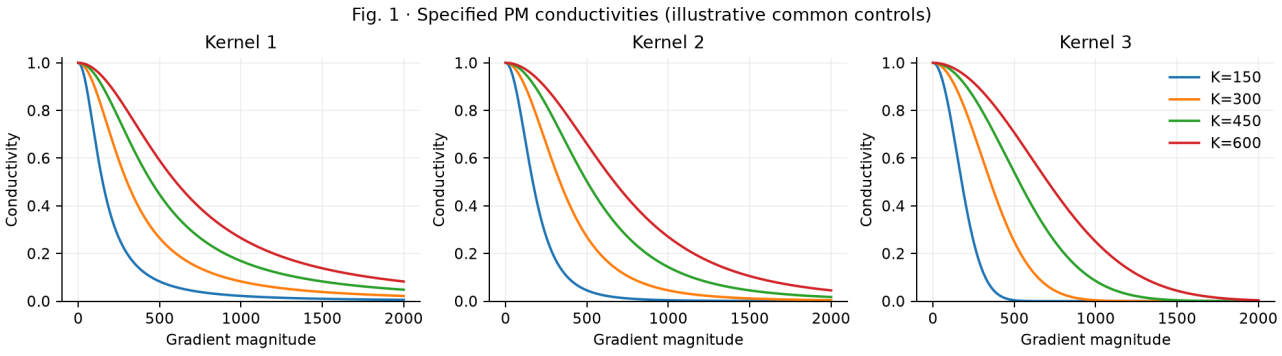
Reference

[1] Maulik, R., & San, O. (2016). *An explicit relaxation filtering framework based upon Perona-Malik anisotropic diffusion for shock capturing and subgrid scale modeling of Burgers turbulence*. **arXiv:1605.04593v1**. Versioned preprint. DOI: 10.48550/arXiv.1605.04593.

Figure gallery

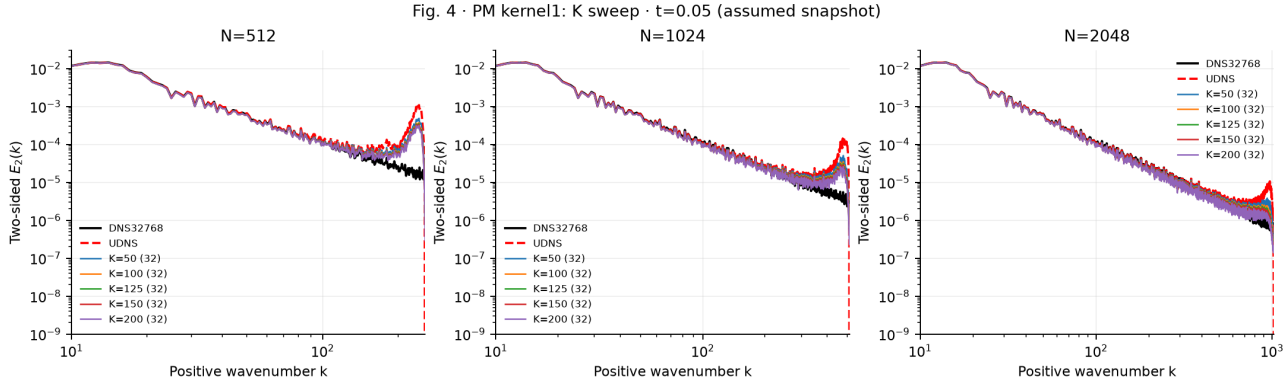
All panels are from this independent reproduction, not digitized author curves. Figures 2 and 3 appear in the results above. Click any image to enlarge it. The gallery preserves the production figures; separate timestep refinements are described in the report.

Figure 1 · Conductivity kernels



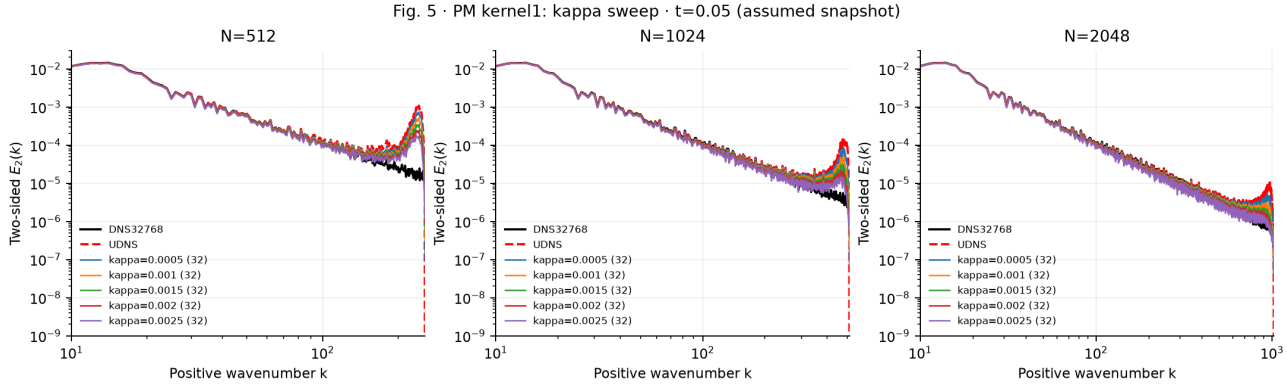
Specified kernel formulas and gradient dependence; an implementation illustration, not an accuracy test.

Figure 4 · PM1 conductivity-scale sweep



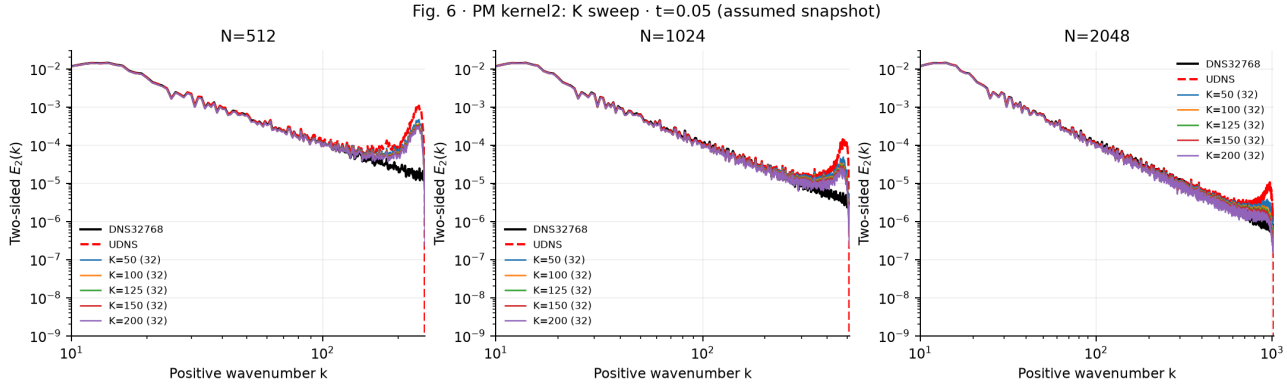
Increasing-control damping at the tested discretizations. The complete useful-SGS gate remains unmet.

Figure 5 · PM1 relaxation-strength sweep



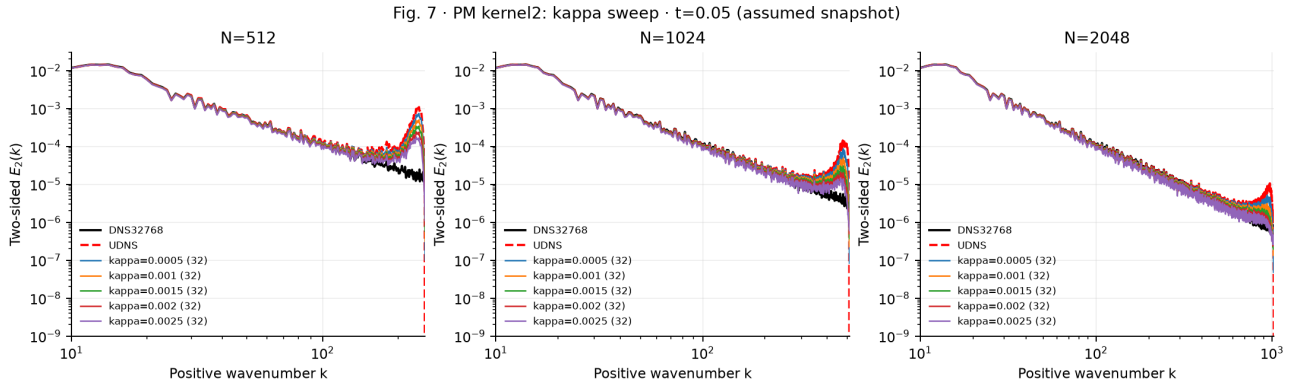
Cutoff damping increases with control strength; this does not establish a globally optimal parameter.

Figure 6 · PM2 conductivity-scale sweep



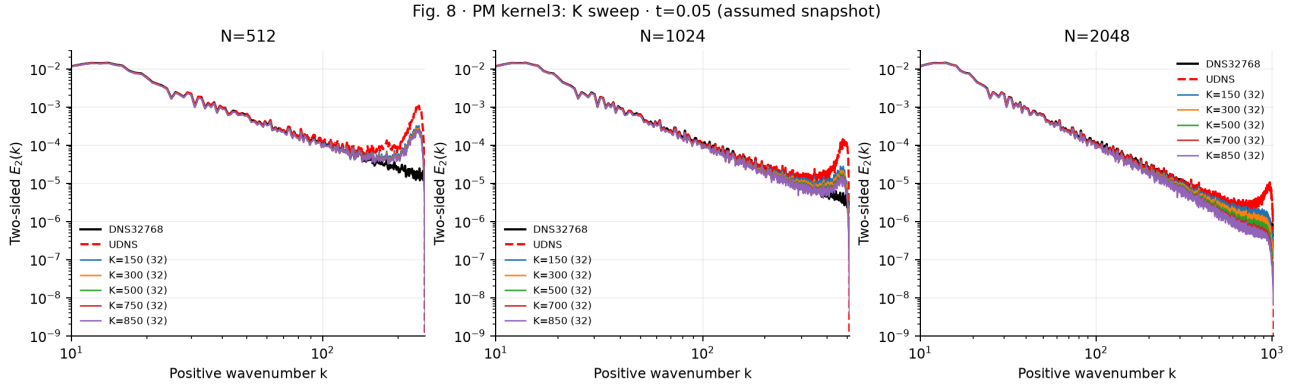
Damping is reproduced. The investigator-defined fixed-band scaling gate remains unmet.

Figure 7 · PM2 relaxation-strength sweep



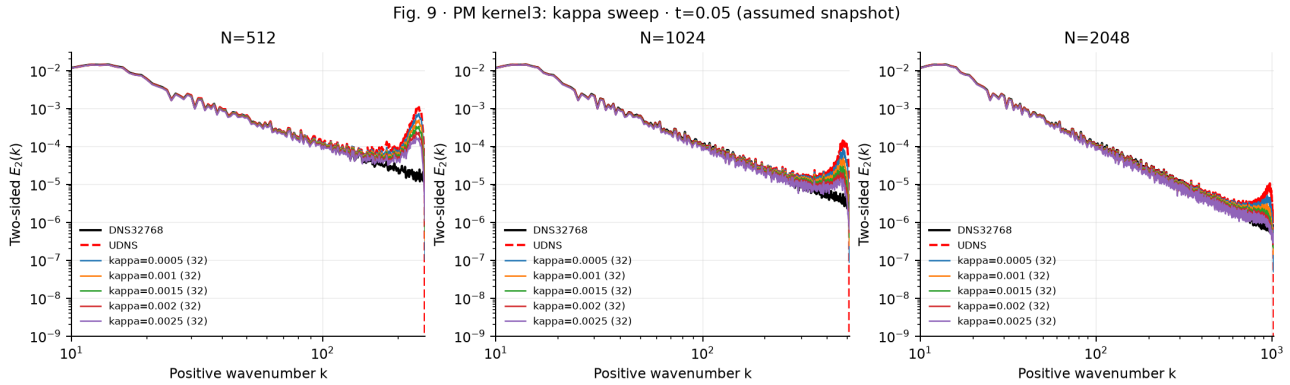
Damping trends retain the production-timestep and low-mode-error qualifications in the report.

Figure 8 · PM3 conductivity-scale sweep



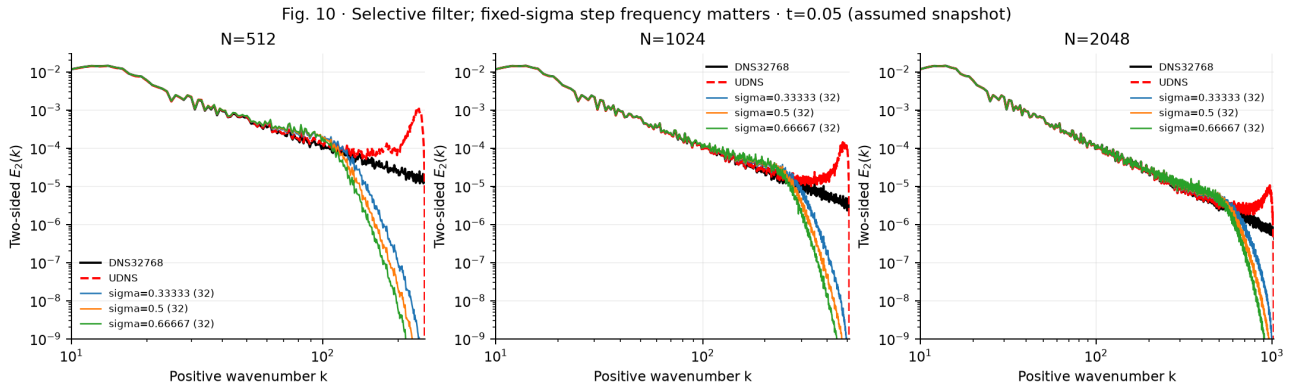
Grid-specific controls are preserved: K=750 at N512, K=700 at N1024 and N2048.

Figure 9 · PM3 relaxation-strength sweep



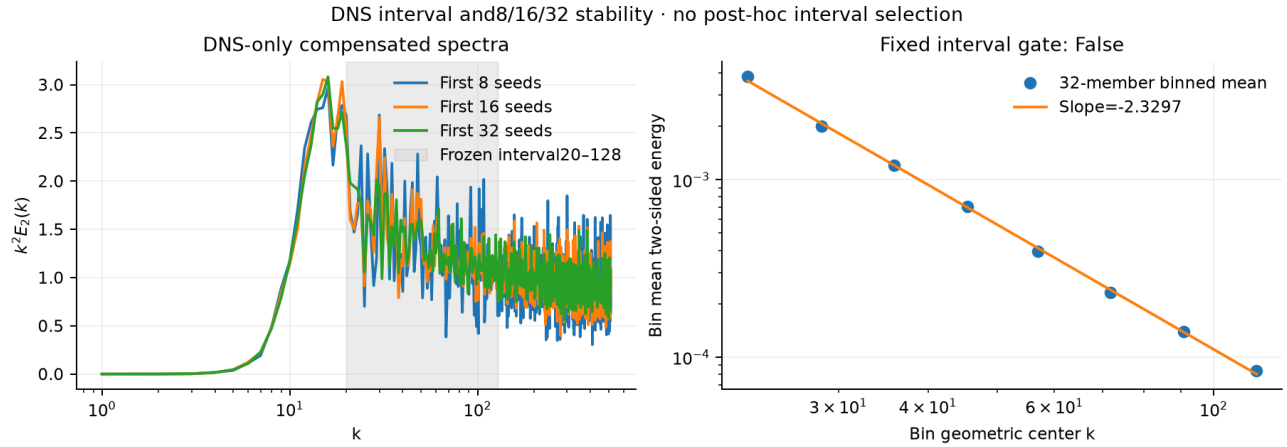
These are production results. A separate finer-level pass does not qualify the original production ensemble.

Figure 10 · Selective-filter sweep



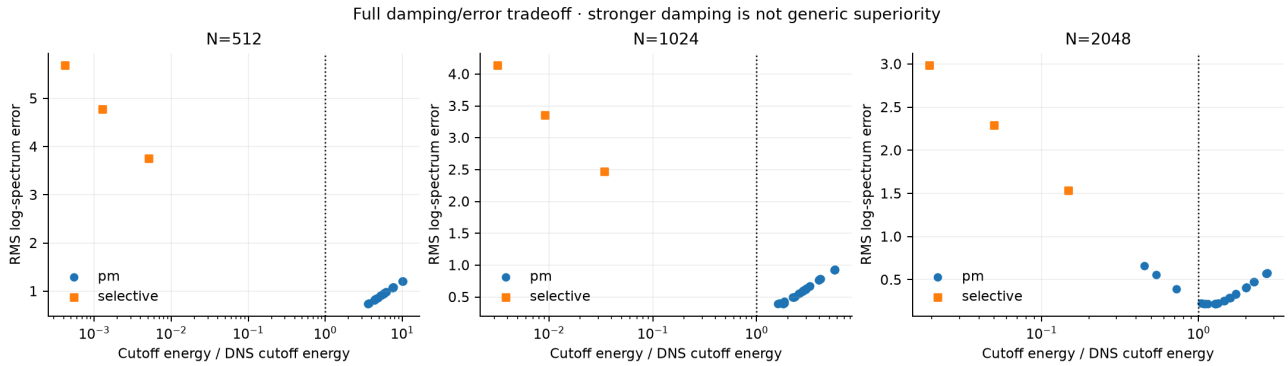
Cutoff suppression is accompanied by spectral deficit. No declared PM/selective pair meets the damping-match criterion.

Supplement · Frozen DNS interval



The fixed 20–128 slope diagnostic fails; it is not a test for the existence of every possible inertial range.

Supplement · Damping–accuracy tradeoff



Cutoff suppression must be read alongside spectral errors and deficits, not as sufficient proof of improved closure accuracy.

HTML presentation of the reviewed Markdown report; no numerical reruns or scientific revisions. Equations use native MathML; every report/gallery figure is embedded. The report reads offline, while linked raw evidence and Markdown documents require the accompanying workspace. External references require internet access.

The review/provenance paragraph is retained as authored. The subsequently completed handoff authorization is in the parent delivery record.

Source SHA-256: `8c56d329c112ad5bfdb7644a4eb0461b607909eff45810ed0350bf9c6899c2c3`